

VFCPL MARINE ENERGY PLATFORM (VFCPL-MEP-001)

# Project SAMUDRA — AMBESS (Anchor Mode BE American Standards (ABS ESS-LiBATTERY) & IR Compliance

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**Applicant & Integrator:** VFCPL (Vayusakti Future Construct Pvt. Ltd.)

**Battery & BMS OEM Partner:** Clean Electric (307.12 Ah LFP Prismatic Cells)

**Domestic Rules:** IRS CN Rev.1 (March 2026) & Guidelines on Battery Powered Vessels (GL Rev.2)

**American Rules:** ABS Requirements for Lithium-ion Batteries (Dec 2025) & Notation ESS-LiBATTERY (Jan 2026)

**Core Architecture:** Direct-Contact Dielectric Immersion-Cooled Pack with Closed-Loop Recirculation

# Applicability Thresholds, Small Battery Rules & Class Notations

| Regulatory Domain            | American Standard (ABS Dec 2025 Guide)                                                                              | Indian Standard (IRS Mar 2026)                                                                        | VF              |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------|
| Mandatory Capacity Threshold | >= 20 kWh (Lowered from 25 kWh in Apr 2024; confirmed Dec 2025). Mandatory application of ABS Part 4, Ch 8 & Guide. | Mandatory for all lithium-ion systems intended for propulsion or essential services; no lower cutoff. | CO<br>co        |
| Small Battery Clause         | 2 to <20 kWh (Section 1/3.1): Specific simplified rules for units in control stations. Not a blanket waiver.        | Handled on plan approval case-by-case; basic IEC 62619 safety compliance required.                    | NO<br>>=<br>ex  |
| Class Notation Assignment    | ESS-LiBATTERY (ABS Jan 2026 p.59). Optional vessel notation verifying compliant installed battery system.           | BATTERY PROP (Main Propulsion) or BATTERY (PROPULSION SUPPORT) (Auxiliary/Hybrid).                    | AL<br>Li<br>SU  |
| US Statutory Framework       | USCG NVIC 02-19, 46 CFR Subchapter J (Electrical), NFPA 855 (Stationary Energy).                                    | DG Shipping Merchant Shipping Rules & IMO MSC.1/Circ.1453 & Circ.1647.                                | AC<br>pr<br>for |

## What Certificate Should a BESS Supplier Seek? (Section 3)

### 1. Product Design Assessment (PDA)

ABS Design Assessment (5-Yr Validity)

- Technical review of defined product against stated rules & standards.
- Review of engineering drawings, SLDs, thermal models, and FMEA.
- Verification of accredited cell test reports (IEC 62619, IEC 62620).
- Verification of thermal runaway propagation evidence (UL 9540A).
- Satisfies technical design review for any shipyard or project.

### 2. Manufacturing Assessment (MA)

ABS Factory & Works Assessment

- Assessment of factory production and quality control processes.
- Applies to Clean Electric (BMS/modules) and VFCPL (pack assembly).
- Requires ISO 9001:2015 certified quality management system.
- Requires annual surveillance plant audits by attending surveyor.
- Guarantees manufactured units reproduce the assessed design.

### 3. Type Approval (TA)

ABS Type Approval

- ABS Type Approval completed to
- Allows off-the-shelf units without repeat
- Unit Certification units at the m
- Generic product installation co

## What Each Standard Proves: IEC 62619, IEC 62620, UN 38.3 & UL (Section 5)

| Standard / Evidence      | Primary Scope & Purpose                                                                                                               | How to Use in VFCPL Project                                                                                                    | Classification |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|
| IEC 62619:2022           | Industrial lithium cell/battery safety (drop, impact, short circuit, thermal abuse). Required by ABS Sec 2/1.2 & IRS CN Rev.1 §1.4.c. | Obtain full test reports identifying tested cell and assembly; verify certificate scope matches supplied 307 Ah product.       | AC             |
| IEC 62620:2014+AMD1:2023 | Industrial cell/battery marking and performance requirements (capacity verification, endurance, discharge rates).                     | Connect declared capacity (307.12 Ah) and test configuration to marketed model. Record the 2023 amendment explicitly.          | AC             |
| UN 38.3 Evidence         | Transport-related battery design testing (T1-T8 altitude, thermal, vibration, shock, short circuit).                                  | Maintain separately from marine installation approval dossier. Do NOT present as proof of complete onboard operational safety. | AC             |
| UL 9540A Results         | Thermal runaway / fire propagation characterization (cell, module, unit, installation levels).                                        | Use as supporting hazard evidence; obtain class agreement on equivalence to specified class test.                              | SU             |

## Thermal-Runaway Evidence: The 7 Review Questions (Section 8)

### The 7 Review Questions to Make Reports Usable

- 1. Cell Construction Authenticity:** Does the tested cell have exactly the same construction as the purchased cell?
- 2. SOC & Conditioning:** Does report identify SOC (100%), conditioning, initiating method, and sample orientation?
- 3. Worst-Case Arrangement:** Was the tested assembly representative of the most challenging offered arrangement?
- 4. Active Systems State:** Were cooling (pump P1) and protective systems running, failed, or unavailable?
- 5. Propagation Containment:** What happened to adjacent cells/modules, enclosure pressure, and gas release?
- 6. Measured Gas/Heat Data:** Are measured gas/heat data sufficient for the integrator's ventilation and fire analysis?
- 7. Invalidation Boundaries:** Which changes in spacing, barriers, enclosure, firmware, or coolant invalidate the inference?

### Class Propagation Rules

#### ABS Section 2/2.1 Mandate:

The battery system must prevent cell-to-cell propagation and demonstrate module-to-module containment. The maximum casing temperature < 100°C.

#### UL 9540 vs. UL 9540A Distinction:

UL distinguishes UL 9540A test methods from UL 9540 certification criteria. A report from one is not directly certified to the other.

#### VFCPL Direct Immersion Advantage

The dielectric fluid conducts heat away from the cell points. In bench tests, zero cell-to-cell

## Liquid & Immersion Cooling: Hydrotest & Failure Detection (Section 2 & 4)

| Engineering Checkpoint       | ABS Dec 2025 Rules (Sec 2/2.4 & 3/2)                                                    | IRS CN Rev.1 / GL Rev.2 Requirements                                          | VF        |
|------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------|
| Shop Fabrication Hydrotest   | 1.5 x Design Pressure for 30 minutes during manufacturing. Zero leakage permitted.      | CN Rev.1 5.2.d.viii: Pressure test to verify pressure-withstanding capacity.  | GA<br>VF  |
| Installation Tightness Test  | Tested at 1.0 x Design Pressure following shipboard installation. Zero leak.            | GL Rev.2 §3.3.3 functional test post-installation.                            | CO<br>pro |
| Pipe Routing Over Live Parts | No mechanical pipe joints directly over live electrical components unless shielded.     | GL Rev.2 2.3.4: Protection against fluid ingress and leakage onto live parts. | CO<br>no  |
| Cooling Failure Alarms       | Loss of coolant flow or high coolant temp must raise immediate alarm at manned station. | CN Rev.1 4.4.k & GL Rev.2 2.3.9.m: Cooling failure alarm mandatory.           | CO<br>De  |

## Dielectric Coolant Fluid: Electrical, Fire & Environmental Properties

### Electrical Insulation

- Dielectric Breakdown: > 50 kV (IEC 60156 / ASTM D877).
- Volume Resistivity: >  $10^{12}$  Ohm-m at 25°C.
- Prevents arcing across live busbars up to 1000 VDC even during dynamic vessel motion.
- Eliminates conductive path hazards common to water-glycol cold plates.

### Thermal & Fire Safety

- Flash Point: > 160°C (ISO 2719 Closed Cup).
- Fire Point: > 190°C (ISO 2592 Open Cup).
- Auto-ignition: > 350°C.
- Kinematic Viscosity: < 35 mm<sup>2</sup>/s at 40°C; < 3000 mm<sup>2</sup>/s at -20°C (ASTM D7042).
- Cold-start pumping validated down to -20°C.

### Marine Environmental

- Biodegradable: > 60 days, OECD 301.
- Aquatic Toxicity: > 100 mg/L (OECD 201, 202).
- Material Compatibility: Compatible with NBR/FKM gaskets.
- Non-hazardous to humans.

## BMS Safety Architecture: Critical Overrides Prohibited (Section 2 & 4)

### ABS BMS Rules (Section 2/5)

- **Critical Overrides Forbidden (Sec 2/5.1):**

No manual override or software bypass is permitted for critical safety shutdowns (overvoltage, undervoltage, overtemperature, short circuit).

- **Physical vs. CAN Status Fallacy (Sec 7):**

A CAN status message indicating "safe" does NOT establish that a physical isolation device actually opened. Auxiliary contact feedback required.

- **Loss of Communication Safe State:**

Communication timeout (>250ms) between BMS Master and Slave CSCs reverts system to an immediate fail-safe trip.

### Clean Electric BMS Implementation

- **100% Cell-Tab Voltage Coverage:**

Direct measurement of every series cell tab; feed BMS.

- **Hardcoded Non-Volatile Lockout:**

Safety threshold tripping is hardcoded in firmware; lockout exists.

- **Dual Independent Isolation:**

Positive and negative poles fitted with fast-acting semiconductor pyro-fuses.

- **Active Insulation Monitoring (IMD):**

Bender IMD integrated: pre-warning alarm at 100 Ohm/V.

# Marine Environmental Qualification: IACS UR E10 (Section 6)

| Test Description     | Standard Reference           | Test Parameters & Levels                                 | Ha  |
|----------------------|------------------------------|----------------------------------------------------------|-----|
| Power Variations     | IACS UR E10 Test 3           | DC +/-20% steady; 0V drop for 5ms; ripple 10%            | BM  |
| Dry Heat & Cold      | IACS UR E10 Tests 4 & 6      | +70°C for 16h operational; -25°C for 2h operational      | Ma  |
| Damp Heat            | IACS UR E10 Test 5           | +55°C at 95% RH (cyclic, 2 cycles of 24 hours)           | All |
| Sinusoidal Vibration | IACS UR E10 Test 7           | 2-13.2 Hz (+/-1.0mm); 13.2-100 Hz (0.7g) on 3 axes       | Im  |
| Inclination Test     | IACS UR E10 Test 8           | Static roll/pitch 22.5°; dynamic roll 22.5° (10s period) | Ve  |
| EMC & Salt Mist      | IACS UR E10 Tests 10 & 15-21 | Radiated/conducted, burst, surge, ESD; 28d salt mist     | BM  |

## Electrical Protection: Fault Current, Pre-charge & Isolation (Section 7)

### Isolation vs. Containment Argument

#### Critical Principle (from Section 7):

Opening a DC contactor cannot remove energy already stored inside electrochemical cells.

#### Independent Systems:

Separate the electrical-isolation argument from the heat/gas-containment argument in your design review.

#### Prospective Fault Current ( $I_{sc}$ ):

Calculated per IEC 61660-1. Due to extremely low internal cell resistance ( $R_i$ ), fault current rise ( $di/dt$ ) is steep.

#### Sub-Millisecond Clearance:

Semiconductor DC pyro-fuses ( $>20$  kA breaking capacity) in series with positive and

### Pre-Charge & Ungrounded IT Bus

#### Pre-Charge Resistor Circuit:

Limits DC link capacitor inrush current to  $<1.0$  A  
interlocked with inverter DC bus voltage ( $>95\%$ )

#### Continuous Insulation Monitoring (IMD):

Floating ungrounded IT DC distribution network  
detector measuring insulation resistance across

**Pre-Warning Alarm Threshold:** 100 Ohm / Volt

**Critical Trip Threshold:** 50 Ohm / Volt with a

#### Selective Protection:

Coordinated selectivity ensures string fuses clear

## Mechanical Enclosure, Ingress Protection & Pressure Relief

### Ingress Protection (IP Code)

- ABS Section 2/2.2 requires minimum IP44 in dedicated battery spaces and IP67 on open weather decks.
- IRS INST-04 recommends minimum IP44.
- VFCPL Monolithic Enclosure is certified to IP67 throughout (dust-tight and protected against water immersion).
- Exceeds standard battery room requirements and

### Thermal Volumetric Breathing

- Dielectric fluid expands with temperature changes (volumetric expansion coefficient  $\beta \sim 0.0007 / ^\circ\text{C}$ ).
- Internal headspace cushion or expansion bellows accommodates fluid expansion across  $-10^\circ\text{C}$  to  $+65^\circ\text{C}$ .
- Prevents internal overpressure during normal cycling without casing deformation.
- Sealed breather with desiccant prevents

### Pressure

- Spring-loaded elastic deformation
- Sized to disperse pressure during worst-case
- Discharge pressure during shipyard exhaust
- Piped directly to personnel enclosure

# Battery Space Design, Ventilation & Hazardous Area Zoning

| Engineering Parameter     | ABS Dec 2025 Rules (Sec 3/3 & 3/5)                                                      | IRS GL Rev.2 Requirements (Sec 2.2 & 2.6)                                                 | VF             |
|---------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------|
| Room Boundary Fire Rating | Machinery Space of Category A. Structural fire protection A-60 towards high risk areas. | Category A space treatment. A-60 boundaries adjacent to accommodation and control spaces. | Sh<br>bu       |
| Normal Ventilation        | Mechanical ventilation maintaining compartment <= 35°C. Independent ducting.            | Independent mechanical ventilation providing minimum 2 Air Changes per Hour (ACH).        | De<br>ne       |
| Emergency Ventilation     | Mandatory high-rate exhaust triggered automatically upon flammable gas detection.       | Minimum 6 Air Changes per Hour (ACH) starting automatically on gas detection.             | Du<br>wi       |
| Hazardous Area Zoning     | Classified per IEC 60079-10-1. Zone 2 where battery off-gas can accumulate.             | Zone 2 classification. Certified safe electrical equipment at least Ex IIC T1.            | Al<br>an<br>Ex |

# Fire Detection, Gas Monitoring & Water-Based Extinguishing

## Gas & Fire Detection Architecture

### Smoke Detection:

Addressable optical smoke detectors complying with IMO FSS Code positioned throughout the compartment.

### Off-Gas Flammable Gas Detectors (LEL):

- At 30% LFL: Audio-visual alarm at manned control station, automatic start of 6 ACH emergency ventilation, and controlled BMS trip.
- At 60% LFL: Total automatic de-energization of all non-Ex electrical circuits in the compartment.

### Thermal Imaging:

Fixed thermographic / IR cameras recommended by IRS GL Rev.2 §2.6.4.2 to monitor external surface temperatures

## Fixed Water-Based Fire Extinguishing

### Mandatory Water-Based Extinguishing:

Fixed water mist or sprinkler system approved (MSC.1/Circ.1165).

### Why Gaseous Systems are Rejected:

Class rules reject purely gaseous systems (CO<sub>2</sub>) for lithium runaway protection because gases store internal heat from burning cells.

### Immersion Protection Synergy:

In the VFCPL pack, internal cell cooling is provided by water mist protects the room structure, cools the external fires

## Packaged & Containerized Marine BESS (ABS Appendix 3 & Section 9)

### Structural & Marine Lifting

- ISO container structure certified to International Convention for Safe Containers (CSC).
- Offshore lifting lugs certified to DNV 2.7-1 / ISO 10855 for marine crane operations.
- Deck tie-down structural calculations verifying survival under severe vessel heave, pitch, and roll accelerations.
- Finite Element Analysis (FEA) verifying deck weld withstand.

### Fire Separation & Boundaries

- Container outer envelope insulated to A-60 fire standard throughout.
- Located away from life-saving appliances (lifeboats, liferafts) and muster stations.
- Clear of accommodation air intakes and hazardous fuel tanks.
- Self-closing doors with limit switches feeding wheelhouse alarms.

### Integrated

- Internal Fire control head
- External Emergency container acco
- Self-contained closing fire da
- Dedicated e release.

# AMBESS Operational Concept: Hotel Power & Essential Services

## Anchor Mode Operating Philosophy

### Duty Profile:

Supplies vessel hotel and auxiliary electrical loads during anchor watch or station keeping, allowing diesel generators to be shut down.

### Generator Optimization:

Generators operate in short, high-efficiency blocks (>75% MCR) to recharge BESS, avoiding low-load (<30%) engine wet-stacking.

### Solar Offset:

Integrated DC/DC solar array offsets auxiliary charging demand in port/anchorage.

### Demonstrated Cycle Life:

Bench validated over 818 cycles with 89.65% capacity retention (SoH) and minimal

## Essential Services & Redundancy

### Auxiliary Role vs. Main Propulsion:

AMBESS is classed as Auxiliary Power / Propulsion. Refer to IACS UR E10, §1.1.3 and ABS Section 4.

### Blackout Prevention Interlocks:

A fault, trip, or short-circuit on the BESS DC bus triggers the tripping of primary shipboard diesel generators.

### Propulsion Redundancy Exemption:

Because AMBESS is dedicated to hotel/auxiliary power, the propulsion shaft line, mandatory dual redundancy, and emergency stop switchboards are NOT required.

# The 10 Engineering Work Packages for the BESS Supplier (Section 7)

| Work Package             | Evidence VFCPL & Clean Electric Must Assemble                                                                    | Decision to Close      |
|--------------------------|------------------------------------------------------------------------------------------------------------------|------------------------|
| 1. Product Definition    | Architecture, model codes, cell OEM, series/parallel configuration, DC voltage window, energy/power ratings.     | Exactly which variants |
| 2. Cell Sourcing         | Full test reports (IEC 62619/62620), lot traceability, storage history, supplier change notices.                 | Which supplier/model   |
| 3. Module Mechanics      | Drawings, connection design, weights, centre of gravity, fastening and service access.                           | Which installation loa |
| 4. Electrical Protection | Prospective fault-current calculation (Isc), fuse/breaker selection, contactor ratings, precharge and isolation. | Can each credited de   |
| 5. BMS Architecture      | Architecture, signal list, fault handling, software release controls and diagnostic coverage, override lock.     | What is the safe resp  |
| 6. Cooling Engineering   | Heat map, thermal model, fluid specification, flow/pressure limits, leakage controls and hydrotest evidence.     | Which faults require c |
| 7. Off-Gas & Fire        | Chemistry-specific test results, release paths, pressure relief and interfaces to protective systems.            | Which protective func  |
| 8. Converter Interface   | Limits, commands, feedback, timeout rules, stop sequence and restart conditions.                                 | Does each end behav    |
| 9. Production Assurance  | Incoming inspection, assembly records, torque control, end-of-line                                               | How is every delivere  |

# Supplier vs. Shipyard Responsibility Boundary Schedule (Section 9)

| Interface Domain           | BESS Supplier Provides (VFCPL & Clean Electric)                                                      | Integrator / Shipyard Provides                                         |
|----------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1. DC Electrical Power     | Voltage/current envelope, terminal details, fault contribution calculation (Isc), fuses, contactors. | Distribution switchboard, cable transits, blackstart                   |
| 2. Immersion Cooling       | Sealed tank, internal dielectric fluid, pump P1, failure alarms, 1.5x hydrotest QA certificate.      | Space ventilation (<1000 CFM), wheelhouse AMS alarm                    |
| 3. Off-Gas & PRV           | Calibrated PRV, outlet geometry, gas volume rate, allowable backpressure data.                       | Dedicated steel exhaust (air intakes), flame arrestor                  |
| 4. Emergency Shutdown      | Hardwired discrete ESD input terminals (24VDC fail-safe), dual contactors, feedback telemetry.       | Remote ESD button, end-to-end trip verification                        |
| 5. Fire Protection         | Immersion fluid containment data, material safety data, post-fire hazard guidelines in manual.       | A-60 structural boundary, fire extinguishers, based fire extinguishers |
| 6. Hazardous Area          | Gas emission data, IP67 enclosure rating, BMS trip signal on gas input.                              | Zone 2 classification, gas detection trips at 30% LEL                  |
| 7. Mechanical & Structural | Wet mass, 3D Center of Gravity, dynamic acceleration load                                            | Deck foundations, structural reinforcement                             |

## Controlled Submission Folders & 4-Tier Evidence States (Section 11)

### The 8 Controlled Submission Folders

- **Folder 1: Basis**

Intended use (AMBESS hotel power), exclusions, rule editions, model family, approval strategy.

- **Folder 2: Design**

Electrical SLD, mechanical drawings, BOM, fluid schematics, PRV sizing calculations.

- **Folder 3: Safety**

Hazard register, FMEA, protective trip matrix, unresolved installation assumptions.

- **Folder 4: Controls**

BMS specs, override lockout statement, CAN map, Software Quality Plan (SQP).

- **Folders 5-8: Thermal Testing, Manufacturing, Delivery**

### The 4-Tier Evidence States (Section 11)

**Avoid a single "compliant" column:**

A design intention, a test result, and class acceptance levels of evidence.

**1. ACCEPTED:**

Formally approved in writing by the classification office.

**2. SUBMITTED:**

Formally delivered to class; currently pending

**3. GAP:**

Rule requirement identified, but technical evidence or action required.

# Complete Dual-Class Compliance Matrix & Engineering Gap Analysis

| Subsystem         | ABS Dec 2025 Guide         | IRS Mar 2026 Rules            | VFCPL-MEP-001 Baseline            |
|-------------------|----------------------------|-------------------------------|-----------------------------------|
| System Threshold  | Mandatory >=20 kWh         | All propulsion/essential BESS | Modular commercial system >20 kW  |
| Cell Safety       | IEC 62619:2022             | IEC 62619:2022 (CN Rev.1)     | LFP Prismatic Cells (307 Ah)      |
| Cell Performance  | IEC 62620:2023             | IEC 62620 (CN Rev.1 §5.3)     | IEC 62620 rated and laser marked  |
| Thermal Runaway   | Non-propagation test       | CN Rev.1 App.1(b)#5 test      | Dielectric immersion cooling bath |
| Shop Hydrotest    | 1.5x design press, 30 min  | 5.2.d.viii pressure test      | Sealed aluminum casing            |
| BMS Override Lock | Overrides forbidden        | 4.4.h override lockout        | Firmware hardcoded lockout        |
| Environmental     | IACS UR E10 (Pt 4, Ch 9)   | IACS UR E10 (CN Rev.1)        | Clean Electric BMS hardware       |
| Enclosure IP      | Min IP44 space / IP67 deck | Min IP44 (GL Rev.2 2.2.8)     | IP67 sealed aluminum enclosure    |
| Ventilation       | Normal + Emergency exhaust | 2 ACH normal / 6 ACH emerg    | HVAC layout specification         |

## Approval Sequence & 10 First-Meeting Decisions (Section 10 & 12)

### The 8-Step Approval Sequence

1. **Freeze Certification Basis:** Rule editions, duty, model family.
2. **Report-by-Report Gap Review:** Match claims to documents.
3. **Agree Test Plan Before Booking Labs:** Lock lab protocols.
4. **Resolve Architecture Risks:** Cooling, isolation, override locks.
5. **Qualify Representative Design:** BOM, HW, FW control.
6. **Complete Works Assessment:** Factory QA audit.
7. **Deliver Serial Dossier:** Unit test records and deviations.
8. **Support Installation:** Shipyard sea trials & commissioning.

### 10 Decisions for First Technical Meeting

1. **Confirm BESS service** (auxiliary-only duty rule)
2. **Package vs. component** certificate scope.
3. **Representative model** and product family limit
4. **Written decision** on existing IEC / ABS witness
5. **Propagation-test setup** and acceptance criteria
6. **Environmental qualification** locations (IACS)
7. **BMS software assessment** and failure simulation
8. **Unit-certification scope** and factory hold points
9. **Installation dependencies** appearing as certificates
10. **Change-notification thresholds** for cells and components